TECHNISCHE UNIVERSITÄT DRESDEN

Fakultät Informatik Institut für Technische Informatik, Professur für Adaptive Dynamische Systeme

Befehlssatz des Von-Neumann Rechners

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1 Abbreviations

n: 8-bit constant
adr: 8-bit address
S: Sign-Flag
Z: Zero-Flag
C: Carry-Flag

2 Transfer-Instructions

Binary	Hex	Mnemonic	Bytes	Flags	Description
0111 1101	7D	MOV A,L	1	-	Move L-reg. to accu
0111 1110	7E	MOV A,M	1	-	Move contents of addr. def. by L-reg. to accu
0111 0111	77	MOV M,A	1	-	Move accu to memory addr. def by L-reg
0011 1110	3E n	MVI A,n	2	-	Move constant n to accu
0011 1010	3A adr	LDA adr	2	-	Move contents of adr to accu
0011 0010	32 adr	STA adr	2	-	Move accu to contents of adr
0110 1111	6F	MOV L,A	1	-	Move accu to L-reg.
0110 1110	6E	MOV L,M	1	-	Move contents of addr. def. by L-reg. to L-reg.
0010 1110	2E n	MVI L,n	2	-	Move constant n to L-reg.
0011 0001	31 n	LXI SP,n	2	-	Move constant n to SP
1111 0101	F5	PUSH A	1	-	Decrement SP, Move accu to stack
1110 0101	E5	PUSH L	1	-	Decrement SP, Move L-reg. to stack
1110 1101	ED	PUSH FL	1	-	Decrement SP, Move flags to stack
1111 0001	F1	POP A	1	-	Move stack to accu, increment SP
1110 0001	E1	POP L	1	-	Move stack to L-reg., increment SP
1111 1101	FD	POP FL	1	-	Move stack to flags, increment SP
1101 1011	DB adr	IN adr	2	-	Move accu to port adr
1101 0011	D3 adr	OUT adr	2	-	Move port adr to accu

3 Arithmetic Instructions

Binary	Hex	Mnemonic	Bytes	Flags	Description
0011 1100	3C	INR A	1	S,Z,C	Increment Accu
0010 1100	2C	INR L	1	-	Increment L-reg.
0011 1101	3D	DCR A	1	S,Z,C	Decrement accu
0010 1101	2D	DCR L	1	-	Decrement L-reg.
1000 0111	87	ADD A	1	S,Z,C	Add accu to accu
1000 0101	85	ADD L	1	S,Z,C	Add L-reg. to accu
1000 0110	86	ADD M	1	S,Z,C	Add memory to accu
1100 0110	C6 n	ADI n	2	S,Z,C	Add n to accu
1001 0111	97	SUB A	1	S,Z,C	Subtract accu from accu
1001 0101	95	SUB L	1	S,Z,C	Subtract L-reg. from accu
1001 0110	96	SUB M	1	S,Z,C	Subtract memory from accu
1101 0110	D6 n	SUI n	2	S,Z,C	Subtract n from accu
1011 1111	BF	CMP A	1	S,Z,C	Compare accu with accu
1011 1101	BD	CMP L	1	S,Z,C	Compare L-reg. with accu
1011 1110	BE	CMP M	1	S,Z,C	Compare memory with accu
1111 1110	FE n	CPI n	2	S,Z,C	Compare n with accu

4 Logic Instructions

Binary	Hex	Mnemonic	Bytes	Flags	Description
1010 0111	A7	ANA A	1	S,Z,0	And accu with accu
1010 0101	A5	ANA L	1	S,Z,0	And L-reg. with accu
1010 0110	A6	ANA M	1	S,Z,0	And memory with accu
1110 0110	E6 n	ANI n	2	S,Z,0	And n with accu
1011 0111	B7	ORA A	1	S,Z,0	Or accu with accu
1011 0101	B5	ORA L	1	S,Z,0	Or L-reg. with accu
1011 0110	B6	ORA M	1	S,Z,0	Or memory with accu
1111 0110	F6 n	ORI n	2	S,Z,0	Or n with accu
1010 1111	AF	XRA A	1	S,Z,0	Exclusiv accu with accu
1010 1101	AD	XRA L	1	S,Z,0	Exclusiv L-reg. with accu
1010 1110	AE	XRA M	1	S,Z,0	Exclusiv memory with accu
1110 1110	EE n	XRI n	2	S,Z,0	Exclusiv n with accu

5 Branch Instructions

Binary	Hex	Mnemonic	Bytes	Flags	Description
1100 0011	C3 adr	JMP adr	2	-	Jump unconditional
1100 1010	CA adr	JZ adr	2	-	Jump on zero
1100 0010	C2 adr	JNZ adr	2	-	Jump on non zero
1101 1010	DA adr	JC adr	2	-	Jump on carry
1101 0010	D2 adr	JNC adr	2	-	Jump on non carry
1100 1101	CD adr	CALL adr	2	-	Call unconditional
1100 1100	CC adr	CZ adr	2	-	Call on zero
1100 0100	C4 adr	CNZ adr	2	-	Call on non zero
1101 1100	DC adr	CC adr	2	-	Call on carry
1101 0100	D4 adr	CNC adr	2	-	Call on non carry
1100 1001	C9	RET	1	-	Return

6 Special Instructions

Binary	Hex	Mnemonic	Bytes	Flags	Description
0111 0110	76	HLT	1	-	Halt
0000 0000	00	NOP	1	-	No operations
1111 1011	FB	EI	1	-	Enable Interrupt
1111 0011	F3	DI	1	-	Disable Interrupt